ALIoT Smart Lab

Engineering dossier - current state, changes, evidence, outstanding work and roadmap

Report version 1.0 · generated 2026-10-18 · system snapshot 2026-10-18 (live portal http://192.168.1.200, laptop host)

GIU BSc thesis project · repository `smart-lab-portal`, branch `offline-router-door-photos` (changes deployed, **not committed**)

> Every result is labelled: **VERIFIED** (measured / tested on the running system or by automated tests), **SIMULATED** (synthetic data, never physical evidence), **PHYSICAL VALIDATION REQUIRED**, **BLOCKED** (a named prerequisite missing), **PARTIAL**. During this snapshot the laptop was **not at the lab**: camera 1, both robots and all ESP32 nodes were unreachable, so no hardware result in this report is a new physical test.

Executive summary

The Smart Lab is a FastAPI + PostgreSQL + React portal running in Docker on the lab laptop, with native Windows services for two security cameras (YOLO26s on the GTX 1650 via DirectML), a face server, a local LLM (Ollama, CPU), a project-knowledge service, read-only robot collectors (UR5e RTDE, FANUC web pages), a PC agent, n8n automations (26 workflows) and ESP32 nodes (door master, ESP32-CAM, consumables bin, reusable bin, environment node).

Software state: VERIFIED. The full automated backend suite, the camera/vision suite, the knowledge, PC-agent and robot-monitor suites, TypeScript checks, production builds and browser end-to-end checks pass (section 7). The final evidence run: 41 PASS, 13 FAIL, 7 NOT AVAILABLE, 3 SIMULATION, 4 PHYSICAL VALIDATION REQUIRED, 1 BLOCKED - every FAIL is a physical device that is not connected here.

Most important changes in this round (all deployed, tested; section 3):

- Data-integrity defect fixed: a resolved camera alert made the next camera event fail (HTTP 500, event and snapshot lost).
- Safety: a latched robot STOP was silently cleared by a backend restart - now rebuilt from the audit log.
- Identity: four defects found with a new simulated scenario bench (stale track stealing a runner's name, velocity overshoot swapping neighbours, door naming by timing alone, cross-camera names never shared). 180 simulated runs now show 0 wrong-name frames.
- Camera protection: one AI answer used every CPU core at above-normal priority and took camera analysis from 7 to 1.7 frames/s; now bounded (4 threads, one answer at a time, cameras above normal priority, instant answers for status questions: ~50 s -> < 0.2 s).
- Reliability centre: alert grouping and quality, dependency model with what-if simulation, investigation, configuration drift, performance objectives, restore drills, action policy, global search, regrouped navigation, safe deploy script with rollback.

Most important risks: (1) the administrator still uses the published seed password (owner action); (2) the laptop is at its memory limit with everything open (commit 95 %) and its clock drifts ~55 ms/h; (3) no physical camera-identity, robot or ESP32 test has been repeated since 14 Oct - the day plan is in `docs/LAB_GO_NOGO_2026-10-18.md`.

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1. Scope and architecture

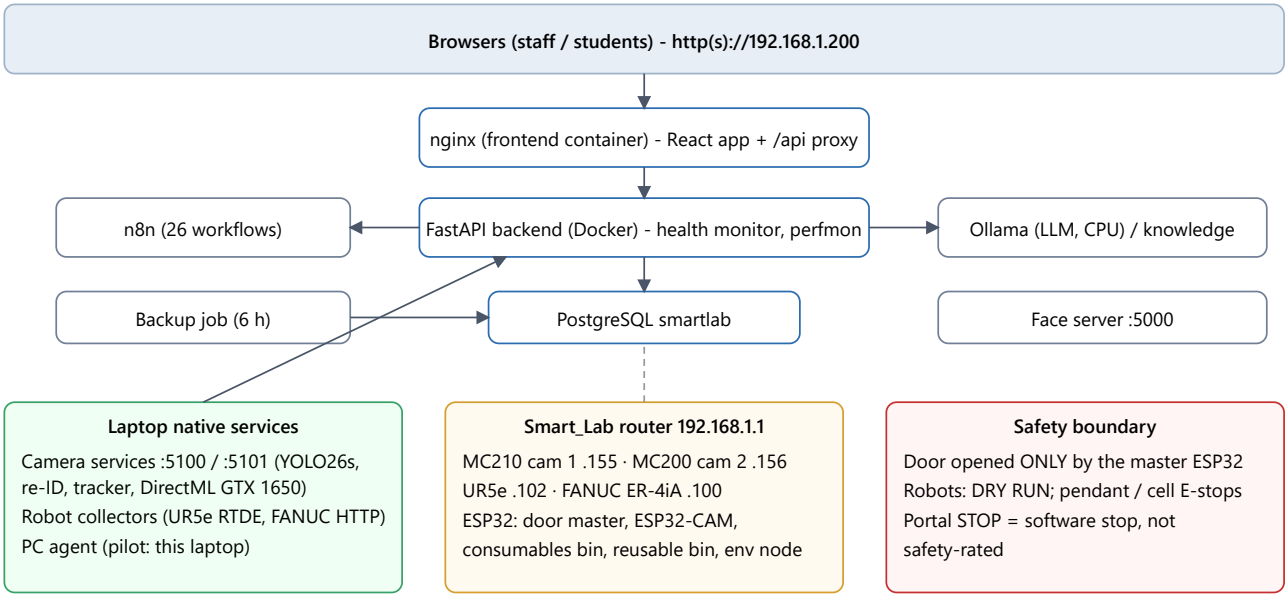


Figure 1 - Deployment and data flows (from docker-compose.yml, launch.bat and the services' code).

Data flows: ESP32 nodes POST heartbeats / telemetry / access events to the backend; camera services post state and events and serve frames that the backend proxies; robot collectors post telemetry (read-only towards the robots); the PC agent posts state and frames and long-polls for viewer / action changes; n8n calls backend endpoints on schedules; the backend serves the portal and the live WebSocket stream.

2. Inventory

2.1 Services (VERIFIED running at the snapshot unless stated)

Component	Location in repo	Interface	State at snapshot
Backend API	`backend/app` (23 routers incl. new reliability, search)	/api, :8000 behind nginx	healthy; deployed version f5eaeff-dirty-20261018144505
PostgreSQL	docker-compose `db`; migrations `backend/alembic` (head d5e8b2c4f917)	internal	healthy, 35 MB
Frontend	`frontend/src` (React/TS/Tailwind, Vite)	:80 / :443 (local CA)	serving; CSP and security headers
n8n	`automation/n8n/json` (repo copies)	:5678	26 workflows, all active; 405 runs / 24 h, 0 failed (read-only check)
Backups	docker-compose `backup`	./backups	every 6 h, verified with pg_restore --list; restore drill VERIFIED
Security camera services	`tools/security_vision`	:5100, :5101	cam 2 streaming 7 fps; cam 1 no picture (camera unreachable)
Face server	`firmware/face_server`	:5000	online
Ollama	native app; portal model qwen2.5:3b, knowledge qwen2.5:1.5b	:11434	online (CPU)

Knowledge service	`tools/knowledge`	:5199 (key)	online (restarted with new settings)
Robot collectors	`tools/robot_monitor`	RTDE / HTTP GET (read-only)	running, no joint data (robots not reachable)
PC agent 0.8	`tools/pc_agent`	HTTPS/HTTP to the API	pilot on this laptop; student PCs need IT approval

2.2 Registered devices (devices table, 13 rows)

Device	Kind	Firmware / agent (reported)	Last report (Cairo)	Status
MASTER_LAB01	door master ESP32	portal-1.0	10-13 02:51	offline - physical check
CAM_LAB01	ESP32-CAM (door)	-	never (device row)	unreachable
BIN_NODE_1	consumables bin	bin-0.2	10-13 02:51	offline
REUSE_BIN_1	reusable bin	reuse-mod-0.1	10-13 02:51	offline
ENV_LAB01	environment node	env-0.1.1	10-11 22:17	offline
SECCAM_LAB01 / 02	security cameras (MC210 / MC200)	-	live	01 no picture · 02 OK
UR5E_01 / FANUC_01	robots (collectors)	robot-collector-0.2	live (collector)	no joint data
UR5E_SIM / FANUC_SIM	simulators	robot-collector-0.2 (sim)	10-08	SIMULATION (never evidence)
PC_ADMIN_LAPTOP	PC agent	pc-agent-0.8	live	online, clock +0.02 s
FACESRV_01	face server record	-	-	checked by the health monitor

Configuration drift check: every reported firmware/agent version matches the repository source (baseline 2026-10-18.1). The three ESP32 nodes went silent in the same minute on 13 Oct, five minutes after the door master rebooted (COMPONENT_FAULT alarm) while the portal kept running: a shared supply or Wi-Fi cause is most likely (correlation, not proof) - to be checked at the bench.

2.3 Data held (exact counts at the snapshot)

Table	Rows	Table	Rows
users	8	sensor_readings	33,012
labs	11	robot_events	8,629
devices	13	audit_logs	3,684
assets	26	notifications	2,273
bookings	23	access_events	1,309
alerts (incidents)	377	security_events	658
issues	4	inventory_transactions	63

Retention: indefinite; no automatic deletion anywhere. Production data was only read during this round (one additive migration).

3. Change register (this round, 17-18 Oct 2026)

Pre-existing functionality (portal pages, access control, incidents lifecycle, reports centre, anomalies, PC screen sharing, robot twins, camera identity pipeline) is retained; items below are new or changed, with their tests.

Area	Change	Where	Tests / evidence	Status
Robots	UR5e Home = pendant program (load+play), readiness checklist, STOP / STOPPED from feedback, conflicting Home refused; FANUC identity + installed options parsed from its own page; FANUC motion blocked with 5 dependencies; STOP latch survives restarts	robot_control.py, robots.py, fanuc_summary.py, RobotControlPanel.tsx	36 backend robot tests; 4 parser tests; e2e robot_readiness (ALL PASS); live restart check	VERIFIED (software); physical BLOCKED / pending
Cameras	Reconnect with exponential backoff + FFmpeg timeouts, stream states, real CPU / memory / GPU metrics, above-normal priority	security_vision.py, latency.py	test_camera_reconnect, test_load_sampler; live test instance (RTSP stream, dead camera states)	VERIFIED on a test instance; active after camera-service restart
Identity	Door naming only for a person first seen at the door after the entry and alone (else review event); posture-change association; staged association; turn handling; crossing re-deal flagged; cross-camera state fix	vision.py, identity.py, crosscam.py	187+ vision tests; scenario bench 180 runs: 0 false names	SIMULATED + unit tests; real accuracy PHYSICAL VALIDATION REQUIRED
Incidents	Grouping by fingerprint (occurrences, last seen, reopen), MITIGATED, escalation on repeats, alert quality, camera "no picture" incident, stale offline incidents cleared; camera-event HTTP 500 fixed	alert_grouping.py, incidents.py, devices.py, security.py; migration d5e8b2c4f917	test_alert_grouping (12), updated security / environment tests	VERIFIED (deployed)
Reliability centre	Briefing, dependency graph + what-if, drift, performance + objectives, alert quality, backups + restore drill, action policy, investigation, global search, regrouped navigation	reliability.py, search.py, dependency_graph.py, investigate.py, drift.py, perfmon.py, briefing.py, policy.py, Reliability.tsx, CommandPalette.tsx	test_reliability (11), authz sweep, e2e reliability.mjs (ALL PASS)	VERIFIED
AI	Instant answers for robot / camera / PC / alert status; 4 threads; one generation at a time; knowledge embed keep-alive 2 h	agent.py, ai.py, config.py, tools/knowledge	test_agent, test_ai_local (incl. concurrency), live timing	VERIFIED

PC fleet	Agent 0.8: host name, own IPv4 (virtual adapters filtered), agent version, clock offset shown	pc_agent.py, equipment.py, NetworkedEquipment.tsx	agent + equipment tests; live on the laptop	VERIFIED (pilot)
Operations	clock_check.py, restore_drill.py, run_evidence.py extended, deploy.py (version, smoke, rollback), /api/health version + db_time	tools/evidence, tools/deploy, main.py	runs recorded in docs/reports	VERIFIED

4. Hardware and wiring (pin assignments from firmware source code)

Pins below are what the firmware **defines**; they are physical wiring only where the bring-up log or the owner verified it (marked ✓). Nothing here was measured during this round. Supply plan (bring-up log): 12 V / 10 A PSU -> master DC switch -> WAGO distribution -> fused branches (Environment, Consumable, Reusable, Door) -> XL4016 buck to 5.00 V per node. Branch fuse values must come from **measured** branch current (BRINGUP_LOG P1.9) - not yet recorded.

Node	Function	GPIO (firmware)	Notes
Door master (SmartLab_Master_Portal)	Lock relay	26	✓ via BC557 PNP; lock on NO = locked when the ESP32 stops driving (fail-secure). Manual egress: check physically.
	PN532 NFC (I2C)	SDA 19, SCL 18	✓ HW-147, 0x24; IRQ/RESET not wired
	Door reed	27	INPUT_PULLUP, LOW = closed
	TFT (SPI)	SCK 13, MOSI 2, CS 25, DC 32, RST 33	MOSI moved off GPIO4 (fingerprint)
	Fingerprint UART	RX 16, TX 17	
	LEDs / buzzer / audio	green 14, red+buzzer 21, red 15, audio 22	audio: PAM8403 via 3.3 k + 100 nF
Environment node	DHT22	4	read every 2.5 s; stale after 15 s
	PIR x2	27, 26	60 s warm-up
	MQ-2	AO 39 (10k/15k divider), DO 25	not a certified smoke detector
	5-channel flame	13, 32, 33, 35, 36	active level to verify per output
Consumables bin	HX711 load cell	DT 32, SCK 33	
	LCD (I2C)	SDA 21, SCL 22	

Reusable bin (modular)	HX711 (5 kg cell)	DT 32, SCK 33	✓ cal -518.04 counts/g vs 200 g reference; accuracy with independent masses NOT checked
	PN532 NFC	SDA 18, SCL 19	✓ (reversed vs the door)
	Lid servo MG90S	25 (signal)	power from external 5 V; max hold 15 s
	Reed (lid)	27	pull-up
	LCD 16x2 (I2C)	SDA 21, SCL 22	pull-up voltage on 5 V backpack NOT measured (risk to GPIO)
ESP32-CAM (door)	OV2640, AI Thinker	board pins	VGA JPEG, CAMERA_GRAB_LATEST, frame buffers in PSRAM; results with age so stale ones are ignored

Review items: LCD backpack pull-ups at 5 V on 3.3 V GPIO (measure); fuse ratings unrecorded; MQ-2 / flame are indicators, a certified detector must stay the authority; ESP32 nodes have no RTC and no offline queue (data during a Wi-Fi gap is lost).

5. Portal guide and navigation map

The staff/admin sidebar is grouped by purpose (no page removed; deep links unchanged). Ctrl+K opens search across pages and devices, incidents, equipment and labs (navigation only - no action is started from search).

Group	Page (route)	Purpose	Roles
Overview	System overview / Operations (/), Laboratories (/labs, /labs/:id)	lab state at a glance, lab pages with equipment galleries	all (content by role)
Live lab	IoT Dashboard (/command), Live monitor (/monitor, own window), Security camera (/admin/security), Robots (/robots, /robots/:id), Operations Center (/admin/operations)	live sensors, camera views (split), robot twins + master controls (DRY RUN)	staff
Safety & access	Access & audit (/admin/access), Bookings (/admin/bookings), Emergency (/admin/emergency)	door events, booking approvals, emergency broadcast	staff / admin
Equipment	Lab Equipment (/admin/equipment: robots, PCs, inventory), System Devices (/admin/devices), Maintenance (/issues)	PC preview / sharing / visible lock, inventory, device registry, tickets	staff
Incidents & alerts	Incidents (/admin/incidents), Alerts (/admin/alerts), Notifications	grouped incidents, lifecycle, investigation	staff
Insight & reports	System analytics (/admin/system), Reliability (/admin/reliability), Reports (/admin/reports)	health, drift, performance, backups, policy; reports centre + PDF	staff / admin
Administration	Users & roles, Settings	accounts, data mode, settings	admin
Students	Dashboard, Laboratories, New booking, My bookings (+ QR), Robots, My reports, Notifications, Profile, Share (/share)	booking, entry QR, viewing a shared screen	students

Other routes kept: /welcome, /sessions/:id, /equipment/:id, /issues/new, /issues/:id, /admin/experiments, /demo, /me, redirects /inventory, /admin/assets, /admin/access-events. Backend authorisation is authoritative (route sweep test covers every API route).

6. AI and camera performance dossier (measured 2026-10-18 on the laptop: i7 6C/12T, 15.8 GB RAM, GTX 1650 4 GB)

6.1 Live-lab AI (portal assistant)

Question type	Before	After	How
"Which robots are online?" / "any alerts?" / cameras / PCs	~48-50 s (LLM with full snapshot, CPU)	**44-142 ms warm, 0.8 s first call**	exact answer from device records (no model)
Lab status, occupancy, reports, bookings	30-100 ms	50-690 ms	existing instant answers
Open questions (LLM qwen2.5:3b)	48 s, camera 2 fell to 4.6 fps (min 1.7)	157-231 s, camera 6.6-6.9 fps (min 3.8-6.0)	4 threads, one answer at a time - deliberate trade-off for camera priority

6.2 Project AI (knowledge)

Path	Measured
Cited lookup (no model)	0.21-0.75 s; cached 0.07 s; cold before the keep-alive fix 2.6 s
Written answer (qwen2.5:1.5b, k=3)	sources at 0.3 s, first text 14-18 s, total 18-31 s under camera load (idle 17 Oct: 4.4 s / 7.5 s); repeat 0.06 s (cache)

Option for the owner: moving the 1.5B model to the GPU would cut written answers to a few seconds but shares the GPU with camera analysis (rule today: LLMs stay on the CPU) - measure first.

6.3 Cameras

Camera	Hardware / stream	Measured	Status
Security camera 2 (SECCAM_LAB02)	Mercusys MC200, 192.168.1.156, RTSP stream2 1280x720	capture 15 fps, analysis 6.8-7.0 /s, detect p50 56-61 ms, frame age p50 37 / p95 98 ms, served frame age p50 104 ms, 0 reconnects	VERIFIED (at home)
Security camera 1 (SECCAM_LAB01)	MC210, 192.168.1.155	unreachable from here (694 open attempts; new code backs off to 30 s)	PHYSICAL VALIDATION REQUIRED
Two pipelines together	cam 2 + a 2nd pipeline on a lab recording	6.95 + 6.97 /s, pass p95 100 / 138 ms, GPU 36 %, 1083 MB of 4096	VERIFIED (2nd source simulated)
ESP32-CAM (door)	AI Thinker OV2640, VGA JPEG	not reachable (ConnectTimeout)	PHYSICAL VALIDATION REQUIRED

6.4 Identity regression (SIMULATED)

Noise level	Runs	Wrong-name frames	Name switches	Track splits
normal (re-ID cos ~0.9, 4 % misses)	60	0	0	1
hard (~0.7, 10 %)	60	0	0	8
harder (~0.5, 15 %)	60	0	0	33

Before the fixes the same bench produced 331 (running) and 276 (side by side) wrong-name frames under hard noise. IDF1 0.95-0.998 (normal). HOTA not computed. Real-world accuracy is not claimed.

6.5 Whole system under load

Scenario	Result
Single user, 10 main reads	before p50 36 / p95 99 ms; **after final deploy p50 31 / p95 80 ms** (825 requests, 0 errors)
10 users (3 s think) + 2 live viewers	before: API p50 121 / p95 376 ms, frames p50 53 / p95 178 ms; **after final deploy: API p50 65 / p95 271 ms, frames (11.7/s) p50 44 / p95 91 ms**; 0 errors (docs/reports/perf_final_20261018.json)
12 users, no think time (stress)	~21 requests/s capacity (one backend process); p95 1.1-1.9 s; 0 errors
10 min sustained, everything incl. LLM before the fix	CPU 99-100 %, RAM 94-98 %; camera 2 dropped to 0.5-3 fps; network 0 % loss (router / camera ping avg 3-4 ms) - led to the AI bounds
Memory growth over 10 min	backend +14 MB (warm-up), camera test instance flat ~1240 MB, PC agent flat ~420 MB: no leak observed

7. Test and acceptance matrix

ID	Suite / check	Command	Result	Date / env
T1	Backend full suite	`docker compose run ... pytest -q`	529 passed, 0 failed (18 min)	2026-10-18, smartlab_test (isolated)
T1b	Safe deploy (tests, schema, build, readiness, 7 smoke checks)	`py -I tools/deploy/deploy.py`	DEPLOYED f5eaeff-dirty-20261018144505, no rollback needed	2026-10-18 14:47 live
T2	Camera / vision suite	`py -3.14 -m pytest tools/security_vision`	187 passed (+30 bench/sampler)	2026-10-18 laptop
T3	Knowledge suite	`py -m pytest tools/knowledge`	40 passed	2026-10-18
T4	PC agent	`py -m pytest tools/pc_agent`	5 passed	2026-10-18
T5	FANUC summary parser	`py -m pytest tools/robot_monitor/test_fanuc_summary.py`	4 passed (incl. the real saved page)	2026-10-18
T6	Frontend types + build	`npx tsc --noEmit`, `npx vite build`	pass	2026-10-18
T7	e2e robot panels	`node e2e/robot_readiness.mjs`	16/16 PASS	live portal
T8	e2e reliability, investigation, search, navigation	`node e2e/reliability.mjs`	ALL PASS (21 menu entries, 7 tabs)	live portal
T9	Concurrency + controls (SIM robot, dry run)	reliability_live.py	360 req, 0 errors; conflict 409; duplicate click DUPLICATE; STOP priority; audit rows	live
T10	Backend restart recovery	docker compose restart backend	ready 26.6 s; STOP latch kept; PC agent back by itself	live
T11	Restore drill + migration	`tools/evidence/restore_drill.py --migrate-test`	restore 5.4-5.7 s, 42 tables; up/down/up OK	isolated DB
T12	Clock agreement	`tools/evidence/clock_check.py`	local clocks within 53 ms laptop 1.3-1.9 s behind internet time	live
T13	Evidence run	`tools/evidence/run_evidence.py --runs 20`	41 PASS · 13 FAIL (devices not connected) · 7 N/A · 3 SIM · 4 PHYS · 1 BLOCKED	live
T14	Frontend dependency audit	`npm audit --omit=dev`	0 vulnerabilities	2026-10-18

T15	Python dependency audit	pip-audit	NOT AVAILABLE (needs install approval)	-

Physical tests (outstanding - none executed in this round)

Test	Expected	Status
Camera 2 mounting, floor calibration (worst fit < 0.15 m), identity scenarios 1-10	no wrong names; Unknown when unsure	PHYSICAL VALIDATION REQUIRED
UR5e commissioning steps 1-10 (docs/ROBOT_CONTROL.md)	Home reached from feedback; STOP latched; nothing restarts	PHYSICAL VALIDATION REQUIRED
FANUC remote interface decision + HOME TP + stop	-	BLOCKED
ESP32 nodes: supply, fuses, voltages, sensors, reconnect after router outage	nodes report; safe output states after power loss	PHYSICAL VALIDATION REQUIRED
Student-PC agent pilot	visible preview, broadcast, lock, audit	BLOCKED (IT approval)

8. Reliability, security and recovery

- **Alert grouping:** fingerprint per condition; repeats counted; reopen within 30 min; MITIGATED; escalation (unacknowledged CRITICAL 15 min / WARNING 4 h, 20 repeats, 3 reopenings). Historical analysis: 376 incidents were only 75 distinct conditions (80 % repeats) - the noise grouping now removes. Safety alarms still notify on every occurrence.
- **Dependency model / what-if:** live states per part with evidence class (live / inferred / simulated); likely roots; simulated failure impact and recovery order.
- **Configuration drift:** baseline `backend/app/config_baseline.json`; compliant at the snapshot; secrets checked only for presence / non-default.
- **Performance:** per-route p50/p95/p99, regression detection (minimum samples, cooldown), objectives with error budget.
- **Backups:** 6-hourly, integrity-checked; restore drill into isolated databases with table comparison; deploy tool with rollback.
- **Security:** strong headers, sign-in lockout, authorisation on every route, audit trail. Open: **admin on the published seed password (High)**; HTTP on the LAN (Medium). Details: docs/SECURITY_FINDINGS_2026-10-18.md.
- **Action policy:** robot motion, door logic, firmware, safety settings, production data: never automatic. The portal STOP is not a safety-rated emergency stop.

9. Procedures

Start-up, shut-down, deploy, recovery, backup and restore are in `docs/OPERATIONS.md` (the runbook); the lab day plan in `docs/LAB_GO_NOGO_2026-10-18.md`; robot commissioning in `docs/ROBOT_CONTROL.md`; node bring-up in `docs/BRINGUP_LOG.md`. Deploy: `py -I tools/deploy/deploy.py` (tests, schema check, rollback point, version stamp, readiness, smoke tests, automatic rollback). Restore drill: `py -I tools/evidence/restore_drill.py`. Evidence: `py -I tools/evidence/run_evidence.py --runs 20`.

10. Outstanding items register

#	Item	Why not done	Who / next step
1	Change the admin password	credentials change needs the owner	owner: Profile > password; tests via environment variables
2	Physical camera, robot, ESP32 tests	laptop not at the lab; devices unreachable	owner + operator: LAB_GO_NOGO sections 1-5
3	Restart the camera services with the new code	restarting active monitoring needs the owner	`launch.bat` at the lab start
4	Laptop clock sync, memory headroom	OS settings / closing apps are the owner's	Settings > Time > Sync now; close unused apps; optional WSL cap
5	ESP32 offline queue + timestamps	needs firmware change and bench flashing	implement + flash with approval (RTC module helps)
6	FANUC interface	needs the qualified operator	choose RSR/PNS or KAREL socket server
7	Student-PC pilot, browser-tab visibility	IT approval / managed extension	request from university IT
8	HTTP to HTTPS redirect	CA must be trusted on lab browsers first	import deploy/tls/smartlab-ca.crt, then redirect
9	Commit the work	commits need the owner's go	review git status; commit on request

11. Roadmap and hardware recommendations

Immediate (days)

- Admin password; camera-service restart; clock sync; lab day tests (identity scenarios with the scoring metrics of the bench).
- Calibrate both floor maps so names can cross cameras; record real ID-switch counts.

Medium term (weeks)

- ESP32 firmware: bounded offline queue with event ids and capture time, idempotent batch ingestion (backend), RTC.
- Serve camera frames directly from nginx (or MJPEG) so live viewers do not load the API process.
- Move Docker services to a small dedicated mini PC: the laptop is at its memory limit with cameras + AI + dev tools.
- Per-branch power monitoring (INA219) feeding the portal: detect a failing supply before nodes go silent together.

Research ideas (thesis follow-up)

- Learned re-ID fine-tuned on consented lab data (needs privacy approval); evaluation corpus with ground truth for IDF1 / HOTA.
- Predictive maintenance only after enough labelled outcomes exist (today: rules + robust baselines).

Hardware purchasing recommendations (indicative prices - retail snapshots, not quotes)

Item	Problem addressed	Priority	Indicative price	Source (seen 2026-10-18)
Small UPS 600-650 VA for laptop, router, 12 V PSU	the 13 Oct same-minute outage pattern; clean shutdown	Recommended	~US$60-90 (US retail)	Staples / Saitech listings via web search
Mini PC (Intel N100, 16 GB) as dedicated server	laptop RAM commit 95 %; services independent of the dev laptop	Recommended	~EUR 189-255; ~AUD 390-480	Notebookcheck (Aoostar N100), Evotech listings
INA219 current sensor per fused branch	branch current for fuse sizing; supply health in the portal	Recommended	~US$7-11 each	Walmart / Electrokit / Hubtronics listings
DS3231 RTC for ESP32 nodes	real capture timestamps for buffered data	Optional (with the offline queue)	~US$5-8 each	Walmart / Adafruit listings
Wired camera (PoE) or Ethernet for camera 1	Wi-Fi cameras drop; RTSP over cable is steadier	Optional	not sourced	-
Certified smoke detector (if not already in the room)	MQ-2 / flame boards are indicators only	Essential if missing (building safety)	not sourced	-

Sources: Staples APC 600VA, Saitech BE600M1, Notebookcheck N100, Evotech N100, INA219 listing, Electrokit INA219, DS3231 listings. Prices vary by region (Egyptian prices not sourced); confirm with a local supplier.

Simplify, do not grow: no new node is needed for the thesis claims; the highest value is finishing the physical validation and the ESP32 offline queue, not more features.

12. Glossary, references, evidence index

Term	Meaning
DRY RUN	robot command evaluated and logged, nothing sent (ROBOT_CONTROL=dry_run)
Fingerprint	stable key of a continuing condition (device | lab | title, or a source key)
IDF1	identity F1 score of a tracker against ground truth
Drift	difference between running configuration and the approved baseline
RTDE	UR real-time data exchange (read-only use here)

Evidence index (machine-readable copy: `docs/reports/dossier/evidence_index.json`):

Claim	Evidence
Final load test after deploy	docs/reports/perf_final_20261018.json; perf_before.json / perf_after*.json
Evidence run summary	docs/reports/evidence_20261018_112213.md / .json
Restore drill + migration test	docs/reports/restore_drill_20261018_024401.json, restore_drill_20261018_140134.json
Identity bench	tools/security_vision/recordings/scenario_bench_*.json (local), test_scenario_bench.py
FANUC identity	tools/robot_monitor/fanuc_probe_out/*SUMMARY.DG.txt (local) + fanuc_summary.py
Robot procedures	docs/ROBOT_CONTROL.md (2026-10-18 section)
Security	docs/SECURITY_FINDINGS_2026-10-18.md
Runbook, backlog, day plan	docs/OPERATIONS.md, docs/BACKLOG.md, docs/LAB_GO_NOGO_2026-10-18.md